

Amber Handal

Evanston, IL (561) 235-4999 amberhandal.dev@gmail.com
github.com/amberhandal linkedin.com/in/amberhandal amberhandal.com

EDUCATION

Northwestern University — Evanston, IL

Sep 2025 – Dec 2026

M.S. in Robotics

University of Florida — Gainesville, FL

May 2019 – May 2024

B.S. in Computer and Information Science Engineering

SKILLS

Reinforcement Learning	PPO, Actor-Critic, Reward Design, Domain Randomization, Sim-to-Real Transfer
Simulation & ML	Isaac Lab, Isaac Gym, MuJoCo, PyTorch, Diffusion Policy, Imitation Learning
Robotics & Control	ROS 2, Cartesian Impedance Control, Inverse Kinematics, Contact Dynamics, State Estimation, SLAM, Motion Planning, Humanoid & Legged Locomotion
Perception	RealSense RGB-D, LiDAR, Elevation Mapping, OpenCV, Sensor Calibration
Software	Python, C++, C, TypeScript, Linux, Git, CI/CD, Docker, AWS

WORK EXPERIENCE

Reinforcement Learning Engineer Intern Persona AI — Pensacola, FL

Jun 2026 – Sep 2026

- Built the company's first perceptive locomotion policies using height-scan-conditioned PPO (Isaac Lab) for complex humanoid locomotion tasks, trained on terrain curricula mirroring the customer worksite.
- Transferred the policies from the Unitree G1 to the company's proprietary humanoid, retuning the observations, action scaling, and rewards for its kinematics, actuator dynamics, and sensor placement.
- Designed the training recipe: asymmetric critic, symmetry augmentation, observation noise, and domain randomization, fine-tuned from a flat-walk policy, plus a supervised base-velocity estimator.
- Collaborated with the perception team to integrate and bring up the sensor stack; specified the height-scan contract, extrinsic calibration, and the flat-floor, box, and shadow-mode test gates.
- Traced on-robot failures from mcap logs of the exact policy inputs to self-body phantoms, uncompensated tilt, and stale cells, and wrote an offline scan-health scorer that gates policy activation.
- Narrowed the sim-to-real gap with a hardware-derived height-scan corruption model ramped by curriculum.

Software Engineer Infotech — Gainesville, FL

Sep 2022 – Aug 2025

- Built automated PDF pipelines (Python, TypeScript, AWS Lambda) to retrieve and classify plan data.
- Mentored three engineers to extend the system for DOT clients using token-based authentication and S3 APIs.

PROJECTS

Soft-Wrist Manipulation — Franka + HSA Stewart Wrist, MurpheyLab

Jun 2026 – Present

- Build the learning stack for a 7-DoF Franka with a soft Stewart wrist: Cartesian impedance control via Polymetis at 1 kHz, SpaceMouse teleop, latency-aligned recording, and diffusion-policy deployment at 10 Hz.
- Design a coarse-fine arm-wrist allocator (complementary filter): fast end-effector error goes to the compliant wrist, the arm recenters it, and the split uses commanded deflection so contact cannot bias the arm.
- Fed the wrist's learned 6-axis wrench estimate into diffusion policy observations for manipulation tasks.

WatchDog — Autonomous Quadruped Inspection

Jan 2026 – Mar 2026

- Built autonomous Go2 inspection with a C++ frontier explorer, Nav2, and a cmd_vel-to-Sport-API bridge.
- Tuned RTAB-Map point-to-plane ICP SLAM using LiDAR, with a PCL ground-filter node for occupancy grids.
- Resolved multi-machine ROS 2 issues: CycloneDDS discovery, QoS tuning, and TF sync via message restamping.
- Integrated SAM 3 on external GPU, projecting segmentation masks to 3D for object change detection.

EKF SLAM from Scratch — TurtleBot3 Localization & Mapping

Jan 2026 – Mar 2026

- Engineered Extended Kalman Filter SLAM from scratch in C++ for TurtleBot3 using ROS 2 and Armadillo.
- Constructed EKF predict/correct cycle with Jacobian linearization for joint robot-landmark state estimation.
- Architected landmark detection with LiDAR clustering, SVD circle fitting, and inscribed angle classification.

PenPal — Vision-Guided Conversational Writing Robot

Dec 2025

- Co-developed a ROS 2 system for a 7-DoF Franka arm to read and write on a moving whiteboard with 3 peers.
- Led perception using RealSense RGB-D, AprilTags, and OpenCV to estimate board pose and publish TF.
- Integrated a Gemini VLM for OCR + response generation via a custom ROS 2 service.
- Developed Cartesian writing execution and generated font-based toolpaths for 3D trajectories using MoveIt.